

Bit Operation Cost of Hold-out Key-Recovery Attacks Against Classic McEliece

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Abstract. We present a self-contained conditional arithmetic model and algorithmic specification for a prospective key-recovery attack on binary Goppa codes, combining the binary hold-out operator with heterogeneous Hasse multiplicities, Lucas-minimal derivative levels, augmented binary block Wiedemann, and reconstruction from local flags. The first two reductions give exact, dimension-guaranteed attack-parameter configurations for every Classic McEliece parameter set. We derive the sparse operator, safe-rank-bound sequence state, a relation-generation tally assuming one attempt per relation batch, higher-flag arithmetic, and the downstream solve performed for every guess. At the selected configurations, modeled affine normalization with dense solves ranges from $2^{172.13}$ to $2^{235.09}$ gates, while projective normalization with nested solves ranges from $2^{146.29}$ to $2^{207.83}$. For `mceliece348864`, a configuration with $c = 7$ gives $2^{142.15}$ modeled gates and retains $2^{61.7}$ bits, excluding temporary solver memory. An exhaustive scan of 19,338 admitted singleton (one-position hold-out) configurations finds a $2^{114.35}$ relation-generation floor in the same fixed model; no downstream-only improvement can cross it. A synthetic experiment heuristically supports the Frobenius-phase balance test, but no public pure cross-pairing is known. None of the tallies is an established break: reliable small-field Krylov yield, higher-flag recovery from the priced truncated block, binary reconstruction, cross-anchor independence, and memory-aware implementation remain open. This manuscript is currently a *living tracking document*.

Keywords: Classic McEliece · Hold-out Key Recovery · Bit Operations · Sparse Linear Algebra

1 Introduction

Classic McEliece derives its public code from a binary Goppa code and relies on the difficulty of recovering an equivalent decoding key [2,10]. Recent higher-order-vanishing and hold-out attacks construct public spaces of multilinear relations that distinguish some Goppa-derived codes from random codes [6,9]. Vedenev subsequently described how suitable hold-out relations could reveal local flags on a hidden coefficient curve and, conjecturally for binary Goppa codes, recover a key [12]. The resulting route is structurally different from information-set decoding: it first computes a large public relation space and then enumerates and interpolates a small number of hidden curve anchors.

This paper consolidates that route into one concrete conditional algorithm and one consistent arithmetic model. Its purpose is accounting, not announcing a completed break. This work has not opened any Classic McEliece key; it has not computed a target-scale operator, kernel, or reconstruction. Some displayed numbers are exact integer consequences of the script-reproduced parameter search, some assume one attempt per relation batch or use leading-monomial models, and all end-to-end totals depend on unproved interfaces. For phase synchronization, we show both the subtotal that omits its execution and the larger subtotal that prices the proposed two-position relation-generation computations; the latter is still conditional because the map they are meant to feed is not constructed.

Our contributions:

1. We explicitly give the binary hold-out operator, strengthen the known Hermite soundness condition by charging the systematic pivots at their full order $d - 1$, and prove new heterogeneous-multiplicity and Lucas-compression statements used to select cells.
2. We give executable-style pseudocode for cell selection, relation supply, first-order certification, sequential higher-flag recovery, flag interpolation, reconstruction, and optional phase synchronization.
3. We specify a nominal Coppersmith block-Wiedemann attempt through an augmented operator whose projected kernel is exactly the required right kernel over $\mathbb{F}_2$, while isolating the missing small-field reliability bound and retry factor.
4. We tally the conditional cost for all five Classic McEliece parameter sets under one attempt per relation batch, including the solve performed once per downstream guess, retained operator state, block-width tradeoffs, higher-flag arithmetic, and every unproved hypothesis.

Artifact. The public artifact repository contains a self-contained costing program `mccost.py`, and concise reproduction instructions for estimates in this paper. See: <https://github.com/mjosaarinen/mccost>

Terminology. The source papers call the public relation space obtained by omitting a column a *hold-out kernel* [6,12]. We call the computation that produces and certifies relations from such a kernel the *relation supplier*, or *supplier* for short. A held-out public column at which the attack seeks a local flag is an *anchor*. An *attack cell*, or simply a *cell*, is a complete choice of shortening, relation degree, held-out set, multiplicities, and retained derivative levels; block width and downstream options turn a cell into a fully priced configuration. A *singleton*, or *one-position hold-out*, has $h = 1$ and supplies one anchor. A *multi-position hold-out* omits $h > 1$ columns; several such held sets form a *cover* if their union contains all required anchors, while a *common hold-out* omits all c anchors at once. A *relation batch* is one block-Wiedemann recovery intended to return at most the declared block width b independent relations; a *relation block* is the verified independent collection assembled from one or more such batches. Appendix A summarizes the status of the ingredients used in the cost tables.

2 McEliece and the Binary Hold-out Operator

Let $q = 2^m$ and let $\Gamma_2 \subseteq \mathbb{F}_2^n$ be the proper binary Goppa code defined by a degree- t irreducible Goppa polynomial and a length- n support over $\mathbb{F}_q$. Its dimension is $k = n - mt$. We use the generator-matrix public-key presentation of McEliece [10]; the Niederreiter formulation used by Classic McEliece is equivalent for key recovery [2]. Public row operations and column permutations put a shortening of length ℓ into systematic form

$$\mathbf{G}_\ell = [I_{k_\ell} \mid R_\ell], \quad n_\ell = n - \ell, \quad k_\ell = k - \ell, \quad n_\ell - k_\ell = mt. \quad (1)$$

Write $p_j \in \mathbb{F}_2^{k_\ell}$ for its columns. The first k_ℓ columns are the weight-one pivots. In Vedenev's parametrization, the non-pivot columns have the hidden form $p_j = y_j F(x_j)$, where $F \in \mathbb{F}_q[Z]^{k_\ell}$ is a coefficient curve of degree at most

$$D_\ell = n_\ell - 2t - 1 = k_\ell + mt - 2t - 1. \quad (2)$$

Let Π_d be the $\mathbb{F}_2$ -space of square-free homogeneous degree- d forms in k_ℓ variables. It has the monomial basis $X_S = \prod_{i \in S} X_i$ indexed by $S \in \binom{[k_\ell]}{d}$, and hence dimension

$$N = \dim \Pi_d = \binom{k_\ell}{d}. \quad (3)$$

Choose a set $\mathcal{H}$ of h held-out non-pivot columns. Each remaining non-pivot column p receives a Hasse vanishing order $s_p \leq d - 1$. For $T \subseteq [k_\ell]$ with $|T| < s_p$, the corresponding row of the binary hold-out operator M is

$$M[(p, T), S] = [T \subseteq S] \prod_{i \in S \setminus T} p_i = [T \subseteq S][S \setminus T \subseteq \text{supp}(p)]. \quad (4)$$

Thus $M\boldsymbol{\eta} = 0$ says that the selected Hasse derivatives of $H = \sum_S \eta_S X_S$ vanish at every constrained public column. Equation (4) is the common binary incidence operator underlying the HOV and hold-out constructions; the two uses differ in their input columns, degree, retained derivative levels, and held-out set [6,9,12].

3 Exact Relation-Generation Reductions

3.1 Hermite Soundness and Heterogeneous Multiplicity

The systematic pivots contribute vanishing for free. A square-free degree- d monomial needs d distinct nonzero coordinates, so at a weight-one pivot every derivative of order less than $d - 1$ vanishes without adding a row to M ; this is the systematic-form optimization of [6,12]. Our Hermite accounting additionally carries that free order- $d - 1$ multiplicity into the interpolation inequality below.

Lemma 1 (Hermite soundness). *Let $H \in \Pi_d$ vanish to order s_p at each constrained non-pivot column. If*

$$k_\ell(d - 1) + \sum_{p \notin \mathcal{H}, p \text{ nonpivot}} s_p > dD_\ell, \quad (5)$$

then $H \circ F$ is the zero polynomial. In particular, H and all consequences of the identity $H \circ F = 0$ remain valid at every held-out column.

Proof. The univariate polynomial $H \circ F$ has degree at most dD_ℓ . Its Hasse zero multiplicities include $d - 1$ at each of the k_ℓ systematic pivots and s_p at each constrained non-pivot node. Distinct nodes with total multiplicity greater than the degree force the polynomial to vanish identically.

Put $r = dD_\ell - k_\ell(d - 1)$ and let $v = mt - h$ be the number of constrained non-pivot columns. The smallest integral order sum admitted by Lemma 1 is $r + 1$. Existing instantiations use one uniform order, which changes this sum in steps of v ; a heterogeneous schedule changes it one unit at a time.

For one public column p assigned order s , its *point block* is the collection of operator rows indexed by all T with $|T| < s$. We call this block *stable-weight* when p has Hamming weight $w \geq d + s - 1$. In this range its exact rank is $\binom{k_\ell}{s-1}$, as shown by (11) below.

Proposition 1 (Balanced heterogeneous schedule). *Assume $2 \leq \lceil (r + 1)/v \rceil \leq d - 1$, that the compared point blocks are in the stable-weight range, and that*

$$d - 2 \leq \frac{k_\ell - \sqrt{k_\ell + 2}}{2}. \quad (6)$$

Define

$$s_{\text{hi}} = \left\lceil \frac{r + 1}{v} \right\rceil, \quad a = r + 1 - v(s_{\text{hi}} - 1). \quad (7)$$

Assign order s_{hi} to a columns and order $s_{\text{hi}} - 1$ to the other $v - a$ columns. This schedule has Hermite slack one. In the stable-weight range, it minimizes the summed point-block rank among all schedules satisfying (5).

Proof. The displayed schedule has total order $r + 1$, hence slack one. In the stable-weight range, the rank charged by order s is $f(s) = \binom{k_\ell}{s-1}$. Writing $j = s - 1$, its second difference satisfies

$$\frac{f(s + 1) - 2f(s) + f(s - 1)}{f(s)} = \frac{(k_\ell - 2j)^2 - k_\ell - 2}{(j + 1)(k_\ell - j + 1)}. \quad (8)$$

Condition (6) makes this nonnegative for every $1 \leq j \leq d - 2$; it also places these indices below $k_\ell/2$, where f is increasing. Hence any schedule with order sum greater than $r + 1$ can be reduced to sum $r + 1$ without increasing its rank charge. If two remaining orders differ by at least two, discrete convexity shows that transferring one unit from the larger to the smaller does not increase their summed charge. Repeating this exchange leaves only the two adjacent orders in (7).

In the tables, $a@s_{\text{hi}} + (v - a)@s_{\text{lo}}$ abbreviates a columns assigned order s_{hi} and $v - a$ columns assigned order s_{lo} .

The proposition is more than a constant-factor optimization. In every cell of Table 2, the largest uniform order whose row-rank bound fits below N is

exactly one below the smallest uniform order satisfying Hermite. For those cells, the uniform feasibility window is empty, while the heterogeneous schedule lands between its two endpoints. This statement does not apply to every later Pareto or sensitivity cell; Table 3, for example, displays an admitted uniform schedule at $(k_\ell, d) = (215, 8)$. Mixed multiplicity enlarges the kernel relative to the high uniform order; semantic validity, rather than equality of kernels, follows from Lemma 1.

3.2 Lucas-Minimal Derivative Levels

Let $W_{i,j}$ denote the binary inclusion matrix from i -subsets to j -subsets. Wilson's decomposition of inclusion matrices and Lucas's parity criterion give

$$W_{i,j}W_{j,d} = \binom{d-i}{j-i} W_{i,d}, \quad (9)$$

$$\binom{d-i}{j-i} \equiv 1 \pmod{2} \iff (j-i) \& \neg(d-i) = 0.$$

We use the resulting binary row-space statement [13]; related modular-rank formulas appear in [1,5].

Theorem 1 (Lucas level compression). *For $0 \leq j < s$, define*

$$\mathcal{U}(d, j) = \{i : 0 \leq i \leq j, (j-i) \& \neg(d-i) = 0\}. \quad (10)$$

If $J \subseteq \{0, \dots, s-1\}$ satisfies $\bigcup_{j \in J} \mathcal{U}(d, j) = \{0, \dots, s-1\}$, then the Hasse rows at levels J span the same binary row space as all levels $0, \dots, s-1$. Consequently, the compression preserves the kernel for a fixed multiplicity schedule.

Proof. The level- j block sees harmonic module $\mathcal{U}_i$ precisely when the coefficient in (9) is odd. The full stack sees every $\mathcal{U}_i$ because level $j = i$ does. The covering condition says that the retained levels see the same modules, and Wilson's diagonal form then gives equality of row spaces.

The minimum-cardinality set $J(d, s)$ is found by a small exhaustive set-cover search over the $s \leq d$ levels. For $(d, s) = (8, 7)$ and $(8, 6)$ it gives $\{0, 4, 6\}$ and $\{0, 4, 5\}$; for $(9, 8)$ and $(9, 7)$ it gives $\{1, 5, 7\}$ and $\{1, 5, 6\}$.

3.3 Rank, Nonzeros, and Cell Selection

At a weight- w point, the exact binary rank of the order- s point block is

$$\rho_{\text{pt}}(k_\ell, w; d, s) = \sum_{b=0}^{s-1} \binom{k_\ell - w}{b} \binom{w}{\min\{s-1-b, d-b, w-d+b\}}, \quad (11)$$

with out-of-range binomial coefficients interpreted as zero. This collapses to $\binom{k_\ell}{s-1}$ once $w \geq d + s - 1$ and can only decrease below that threshold. Therefore

$$\bar{\rho} = \sum_p \binom{k_\ell}{s_p - 1} \quad (12)$$

is an upper bound on the global rank of the stacked operator, and

$$\dim \ker M \geq N - \bar{\rho} \quad (13)$$

is a rigorous dimension guarantee whenever the right-hand side is positive. We neither require nor assert cross-point independence here.

For a point of weight w and retained level set J , the exact number of nonzeros is

$$\nu(w; d, J) = \sum_{j \in J} \sum_{a=0}^j \binom{w}{a} \binom{k_\ell - w}{j - a} \binom{w - a}{d - j}. \quad (14)$$

The total ν is the sum of (14) over constrained columns with their assigned levels. The model uses the uniform-weight convention $w = \lfloor k_\ell/2 \rfloor$ for every non-pivot column and does not optimize which columns receive the lower order. The nonzero count is therefore modeled and weight-dependent even though the cell dimensions and row-rank bound are exact.

We write

$$R_M = \sum_p \sum_{j \in J_p} \binom{k_\ell}{j} \quad (15)$$

for the number of emitted rows after Lucas compression. The local certificate of Algorithm 3 needs $\text{rank } \Psi_1 = k_\ell - (m + 1)$ and then $\text{rank } \Psi_2 = \binom{m}{2}$ on $\ker \Psi_1$. A relation block can support both ranks only if its dimension is at least their sum, so throughout we set

$$r_{\min} = k_\ell - (m + 1) + \binom{m}{2}. \quad (16)$$

This gives the five 64-relation batches at the main `mceliece348864` cells used below. It is only the dimension required by the first-order rank gates; it does not imply that the truncated block determines the sequential higher systems. Assumption 2 isolates that additional interface.

Algorithm 1 combines Proposition 1 and Theorem 1. The scan is exhaustive: nonzero count is not monotone in either k_ℓ or the multiplicity because changing s can replace a deep shadow with a larger but shallower derivative level.

We call a cell *dimension-admitted* when it passes the multiplicity-order gate and Hermite inequality in Algorithm 1 and satisfies $\bar{\rho} < N$. Admission guarantees the positive kernel floor (13); it does not guarantee Krylov yield or downstream recovery.

The selector does not operate on the public key beyond reading its declared shape. For Q candidate triples, minimum-level set cover takes at most 2^d subset tests per multiplicity, and the rank and nonzero sums take $O(d^3)$ exact-integer operations, giving $O(Q(2^d + d^3))$ small combinatorial operations. This one-time model-construction cost is excluded from the cryptanalytic bit tally. The implemented scan is restricted to $h = 1$ and minimizes the complete width-64 singleton supplier objective; cover and common hold-out cases require separately declared richness assumptions and supplier counts.

Algorithm 1: SelectCells() : exact admissibility and modeled supplier price.

Input: (n, m, t) ; finite ranges for (k_ℓ, d, h) ; a declared column-weight model.
Output: All admissible cells with schedules, level sets, dimensions, nonzeros, anchor counts, and resource estimates.

- 1 Initialize an empty list $\mathcal{C}$;
- 2 **foreach** candidate (k_ℓ, d, h) in the declared ranges **do**
- 3 Set $n_\ell = k_\ell + mt$, $D_\ell = n_\ell - 2t - 1$, $v = mt - h$, and $r = dD_\ell - k_\ell(d - 1)$;
- 4 Compute $(s_{\text{hi}}, a, s_{\text{lo}}, v - a)$ from (7); refuse unless $2 \leq s_{\text{hi}} \leq d - 1$;
- 5 Find minimum covers $J_{\text{hi}} = J(d, s_{\text{hi}})$ and $J_{\text{lo}} = J(d, s_{\text{lo}})$ using (10);
- 6 Set $N = \binom{k_\ell}{d}$ and $\bar{p} = a \binom{k_\ell}{s_{\text{hi}}-1} + (v - a) \binom{k_\ell}{s_{\text{lo}}-1}$;
- 7 Refuse if $\bar{p} \geq N$;
- 8 Sum (14) to obtain ν and count the emitted operator rows R_M ;
- 9 Set $U = k_\ell(D_\ell + 1)$, $B = \binom{k_\ell}{2}$, $c = \lceil (U - 1)/B \rceil$, and r_{min} by (16);
- 10 Append the complete record, including the kernel floor $N - \bar{p}$, to $\mathcal{C}$;
- 11 **end**
- 12 **return** $\mathcal{C}$;

4 Matrix-Free Relation Supply

For a retained row (p, T) at level j , matrix application is the restricted $(d - j)$ -step shadow

$$(M\eta)_{p,T} = \bigoplus_{\substack{V \subseteq \text{supp}(p) \setminus T \\ |V|=d-j}} \eta_{T \cup V}. \quad (17)$$

The transpose uses the same traversal with reversed accumulation. We store neither M nor the augmented operator below.

The tempting symmetric operator $M^\top M$ need not have the desired kernel over $\mathbb{F}_2$: if $0 \neq v \in \text{im } M \cap \ker M^\top$ and $Mx = v$, then $x \in \ker(M^\top M) \setminus \ker M$. Candidate verification makes this route sound, but it does not provide adequate yield. We therefore price the exact augmented construction.

Lemma 2 (Exact augmented kernel). *For*

$$\mathcal{A} = \begin{pmatrix} 0 & M^\top \\ M & 0 \end{pmatrix}, \quad (18)$$

one has $\ker \mathcal{A} = \ker M \oplus \ker M^\top$. Projection onto the first block returns exactly $\ker M$.

Proof. The equation $\mathcal{A}(x, y)^\top = 0$ is equivalent to the independent equations $Mx = 0$ and $M^\top y = 0$.

The lemma settles the soundness of the projection, not the output distribution of a finite Krylov run. At the selected $(k_\ell, d, c) = (208, 8, 9)$ cell, which has

the smallest desired-kernel fraction among the main mceliece348864 cells used below, the dimension floors are

$$\dim \ker M \geq 482,845,159,446, \quad \dim \ker M^\top \geq 58,108,396,307. \quad (19)$$

At equality of the row-rank bound, the desired summand is 89.26% of the kernel dimension. This makes catastrophic projection loss implausible but does not prove a yield theorem for a structured small-field Krylov computation. We therefore project, verify, and rank-test every returned vector, and failure to obtain enough relations triggers a restart or refusal.

We price one nominal balanced equal-block Coppersmith block-Wiedemann attempt on $\mathcal{A}$ [3]. Put $D_A = N + R_M$ and $R_A = 2\bar{\rho}$, the safe rank bound for the matrix actually passed to the solver. Choose independent uniform projections $X, Y \in \mathbb{F}_2^{D_A \times b}$, use the identity preconditioner, form $S_i = X^\top \mathcal{A}^i Y$, compute a minimal matrix generator, and perform the standard recovery pass. The retained precision and one-attempt application count are

$$L_b = 2 \left\lceil \frac{R_A}{b} \right\rceil + b, \quad n_A = 3 \left\lceil \frac{R_A}{b} \right\rceil. \quad (20)$$

Let $u = c$ for one-position hold-outs, $u = \lceil c/h \rceil$ for a cover by h -position hold-outs, and $u = 1$ for one common $h = c$ hold-out. A balanced width- b Coppersmith recovery returns at most b relations, so put $a_b = \lceil r_{\min}/b \rceil$. We adopt a sequential full-word model: the a_b recoveries reuse one width- b state, but each sparse incidence is charged as one 64-bit word operation. If r_{BW} attempts are executed per required relation batch, then

$$C_{\text{sup}}(b) = 6 \cdot 64 \left\lceil \frac{2\bar{\rho}}{b} \right\rceil \nu u a_b r_{\text{BW}} \quad \text{binary gates.} \quad (21)$$

Every displayed row sets $r_{\text{BW}} = 1$, meaning one attempt for each of the a_b required batches, not one recovery for the entire supplier. Ideal b -lane charging would replace 64 with b ; packing several width- b instances into one word would instead require a correspondingly wider retained state. Neither alternative is used in the tables. If p_{BW} denotes the unknown per-attempt failure probability, expected work would contain the factor $1/(1 - p_{\text{BW}})$. Coppersmith analyzes block Wiedemann but excludes pathological matrices from the generic projection argument [3]. Eberly's provably reliable block-Lanczos result is over large fields; for small fields he instead uses sparse preconditioning and partial Frobenius decomposition [4]. We supply neither a theorem excluding pathology for these structured binary matrices nor the corresponding small-field conditioning analysis. Thus (21) is a verified-or-refused $r_{\text{BW}} = 1$ tally, not a reliable expected running time.

State has two separately auditable solver terms:

$$S_b^{\text{vec}} = \frac{b(N + R_M)}{8}, \quad S_b^{\text{seq}} = \frac{L_b b^2}{8}, \quad S_b = S_b^{\text{vec}} + S_b^{\text{seq}}. \quad (22)$$

Algorithm 2: SupplyRelations() : matrix-free certified right-kernel supply.

Input: A cell from Algorithm 1, its constrained public columns, block width b , and retry limit $R_{\max}$.
Output: At least $r_{\min}$ independent relations in $\ker M$, or $\perp$.

- 1 Implement **ApplyM** and **ApplyMT** by the shadow traversal (17);
- 2 Define $\text{ApplyA}(x, y) = (\text{ApplyMT}(y), \text{ApplyM}(x))$;
- 3 Initialize an empty certified relation block K' ;
- 4 **for** $j = 1$ **to** $a_b = \lceil r_{\min}/b \rceil$ **do**
- 5 Set $r_j = \min\{b, r_{\min} - \dim K'\}$ and mark this batch unfinished;
- 6 **for** $r_{\text{BW}} = 1$ **to** $R_{\max}$ **do**
- 7 Draw fresh $X, Y \in \mathbb{F}_2^{(N+R_M) \times b}$ and use the identity preconditioner;
- 8 Form $S_i = X^T \mathcal{A}^i Y$ through precision L_b in (20); compute a minimal matrix generator and run Coppersmith recovery;
- 9 Project returned vectors onto the N -coordinate block, verify each by a fresh call to **ApplyM**, and row-reduce them against K' ;
- 10 **if** *at least r_j new independent relations remain* **then**
- 11 Append r_j of them to K' ; mark the batch finished; **break**;
- 12 **end**
- 13 **end**
- 14 **if** *the batch is unfinished* **then**
- 15 **return** $\perp$;
- 16 **end**
- 17 **end**
- 18 Replay the Hermite inequality, schedule, Lucas covers, and all operator annihilations; **return** K' ;

A packed block of $r_{\min}$ recovered relations would occupy $S^{\text{rel}} = r_{\min}N/8$ bytes, but the local jet maps can be accumulated while relations are streamed, so the cap tables report S_b and list neither S^{rel} nor simultaneous copies. Sequence data may likewise be streamed to slower storage only by paying I/O. Recursive generator-basis temporaries, checkpoints, access cost, and replication are omitted, so S_b is neither a peak-RAM measurement nor a complete storage bill.

For direct comparison with the work exponents, both prose and tables report the base-two logarithm of the retained number of bits. Thus a table entry 53 means 2^{53} retained bits.

Algorithm 2 makes the batch loop explicit. One width- b application of M traverses ν incidences and costs 64ν binary gates in the full-word model; one application of $\mathcal{A}$ costs 128ν . Equation (21) follows by multiplying this by n_A , the supplier count, the a_b relation batches, and the attempts per batch. Accepted relations may be repacked into $a_{64} = \lceil r_{\min}/64 \rceil$ full words for fresh annihilation checks. Together with naive independence testing, this adds at most

$$C_{\text{replay}} \leq u (64a_{64}\nu + r_{\min}^2 N) \quad (23)$$

binary gates. We record this as a separately auditable lower-order term rather than silently changing the leading supplier exponent; at every displayed cell it is far below C_{sup} .

For a held set $\mathcal{H}$, write $M_{\mathcal{H}}$ for the operator with those columns omitted and $K_{\mathcal{H}} = \ker M_{\mathcal{H}}$. At a fixed multiplicity assignment on the common constrained columns, $i \in \mathcal{H}$ implies $K_{\{i\}} \subseteq K_{\mathcal{H}}$ because holding out more columns removes rows. The costed h -position cells are then re-scheduled so that (5) stays strict; no containment between the re-scheduled kernels is asserted. A cover of the c anchors by h -position hold-outs reduces the number of relation-supply computations to $\lceil c/h \rceil$, but the containment fact alone does not say that their certified blocks determine the singleton local flags.

Assumption 1 (Multi-position hold-out richness) *For every held set $\mathcal{H}$ in the chosen cover, public restrictions of a certified block from $K_{\mathcal{H}}$ determine the same local flag at each $i \in \mathcal{H}$ as a sufficiently rich block from $K_{\{i\}}$.*

At the $c = 8$, $k_{\ell} = 215$, $d = 8$, $b = 32$ cell, every $h \leq 6$ is dimension-admitted. The direct relation-generation prices in Table 1 show that two disjoint $h = 4$ kernels dominate the proposed two-kernel $h = 6$ cover: they are slightly cheaper, use less state, avoid overlap, and require the weaker enlargement. Every $h > 1$ row is conditional on Assumption 1.

Table 1. Conditional multi-position hold-out covers of the eight anchors at $(k_{\ell}, d, b) = (215, 8, 32)$. Work is $\log_2$ gates; Δ is relative to eight one-position relation-supply computations.

h	Kernels	Kernel floor	Supplier	Δ	$\log_2$ state bits
1	8	6,279,785,122,167	116.71	+0.00	54.00
2	4	5,538,193,278,188	115.71	-0.99	54.01
3	3	4,796,601,434,209	115.31	-1.40	54.02
4	2	4,055,009,590,230	114.73	-1.97	54.03
5	2	3,313,417,746,251	114.74	-1.96	54.04
6	2	2,571,825,902,272	114.75	-1.95	54.05

5 From Relations to a Key

At a held column y , Vedenev’s local construction first separates candidate tangent branches and then solves sequential higher-order systems to obtain a complete nested flag. These are distinct interfaces. For a relation space K' let $\Psi_1(H) = \nabla H(y)$, let $T_0 = \ker \Psi_1^T$, and let Ψ_2 be the quadratic Hasse map induced by $\ker \Psi_1$ on $T_0/\langle y \rangle$. A *branch plane* is the two-dimensional space $\langle y, v_1 \rangle$ determined by one candidate first-order tangent direction v_1 .

A representation-independent but deliberately loose certification bound follows by scanning every degree- d coefficient once for the value, gradient, and

Algorithm 3: CertifyFirstOrder() : first-order branch certification.

Input: A certified relation block $K' \subseteq \ker M$, held column y , and extension degree m .

Output: The m candidate branch planes through y , or $\perp$.

- 1 Check $H(y) = 0$ for every $H \in K'$; otherwise **return** $\perp$;
 - 2 Construct Ψ_1 , put $T_0 = \ker \Psi_1^\top$, and check $\dim T_0 = m + 1$ and $y \in T_0$;
 - 3 Put $K'_0 = \ker \Psi_1$, construct $\Psi_2 : K'_0 \rightarrow \text{Sym}^2(T_0/\langle y \rangle)^*$, and check $\text{rank } \Psi_2 = \binom{m}{2}$;
 - 4 Extract the m proposed pure branches and check exact vanishing on each;
 - 5 Check that coefficientwise squaring acts as one m -cycle;
 - 6 **return** the certified branch planes, otherwise $\perp$;
-

quadratic Hasse maps. For $r_{\min}$ supplied relations,

$$C_{\text{cert}} \leq r_{\min} N \left(1 + d + \binom{d}{2} \right) + O \left(r_{\min} k_\ell^2 + r_{\min} \binom{m}{2}^2 \right) \quad (24)$$

binary operations suffice. This bound is intentionally insensitive to derivative sparsity; it makes the claim that certification is lower-order checkable without relying on a shortcut.

For $H \in K'$ and vectors $v_1, \dots, v_{r-1}$, let $\Phi_{H,r}(y; v_1, \dots, v_{r-1})$ denote the coefficient of ϵ^r in the terms of Hasse order at least two in $H(y + \sum_{j < r} v_j \epsilon^j)$. Vedenev's systems (20)–(21) determine the next vector v_r from

$$\begin{aligned} \nabla H(y) \cdot v_r &= -\Phi_{H,r}(y; v_1, \dots, v_{r-1}) & (H \in K'), \\ \Phi_{H,r+1}(y; v_1, \dots, v_r) &= 0 & (H \in \widehat{K}'), \end{aligned} \quad (25)$$

where $\widehat{K}' = \{H \in K' : \nabla H(y) = 0\}$. The equations are affine in v_r , but their consistency and semantic completeness are not consequences of the first-order rank check.

Assumption 2 (Truncated higher-flag richness) *Conjecture 3 of Vedenev [12] holds for the binary-Goppa relation spaces used here. Moreover, the certified $r_{\min}$ -dimensional truncated block K' is rich enough that every sequential system (25) has the same admissible flag branches as the full hold-out kernel. The number of retained affine branches over all stages is at most the declared multiplier r_{flag} .*

For completeness of the tally, we charge a deliberately loose dense implementation rather than calling this stage free. Rescanning every coefficient to assemble all truncated Hasse expressions and densely solving every k_ℓ -variable system gives the one-time upper model

$$\begin{aligned} C_{\text{flag}} &\leq c m r_{\text{flag}} (C_{\text{asm}} + C_{\text{elim}}) m^2, \\ C_{\text{asm}} &= r_{\min} N d k_\ell^3, \\ C_{\text{elim}} &= (k_\ell - 2) (r_{\min} k_\ell^2 + k_\ell^3), \end{aligned} \quad (26)$$

Algorithm 4: RecoverHigherFlag() : sequential completion of one local flag.

Input: K' , held column y , one certified first-order plane P , and branch limit r_{flag} .

Output: A complete nested flag, or $\perp$.

- 1 Set $v_0 = y$, choose $v_1 \in P \setminus \langle y \rangle$, and set $W^{[j]} = \langle v_0, \dots, v_j \rangle$ for $j = 0, 1$;
- 2 **for** $r = 2$ **to** $k_\ell - 1$ **do**
- 3 Construct both affine systems in (25) from K' and the retained partial branches;
- 4 Solve them over $\mathbb{F}_{2^m}$; substitute every solution into every equation and discard inconsistent branches;
- 5 Quotient solutions by the current $W^{[r-1]}$, retain at most r_{flag} distinct extensions, and refuse on overflow;
- 6 For each retained extension, set $W^{[r]} = W^{[r-1]} + \langle v_r \rangle$ and require $\dim W^{[r]} = r + 1$;
- 7 **end**
- 8 Refuse unless the surviving nested spaces form the unique complete flag associated with P ;
- 9 **return** *that flag*;

binary gates. The initial factor m charges all certified first-order branches at each anchor, and the final m^2 is the same optimistic field-multiplication proxy used below. The tables set $r_{\text{flag}} = 1$. Equation (26) prices arithmetic and possible branch retries multiplicatively; it does not prove Assumption 2. In the displayed cells, it is lower order than the supplier or downstream term.

Suppose Algorithm 4 yields full flags at c held positions, with basis matrices $Q_i \in \text{GL}_{k_\ell}(\mathbb{F}_q)$ and candidate support coordinates $a_i \in \mathbb{P}^1(\mathbb{F}_q)$. Vedenev's Step 3 seeks a polynomial vector in the intersection

$$\mathcal{L} = \bigcap_{i=1}^c Q_i \text{diag}(1, Z - a_i, \dots, (Z - a_i)^{k_\ell - 1}) \mathbb{F}_q[Z]^{k_\ell} \quad (27)$$

with degree at most D_ℓ . It has

$$\begin{aligned} U &= k_\ell(D_\ell + 1) \quad \text{coefficient unknowns,} \quad B = \binom{k_\ell}{2} \quad \text{conditions per anchor,} \\ c &= \left\lceil \frac{U - 1}{B} \right\rceil \quad \text{required anchors.} \end{aligned} \quad (28)$$

The last formula assumes that conditions from distinct positions have their generic incremental ranks. Vedenev states this assumption and expects a one-dimensional correct solution [12]; it is not proved for binary Goppa flags. The identity $B = \binom{k_\ell}{2}$ is also the condition count in Vedenev's Remark 6.

Proposition 2 (Flag-information ceiling). *If the public datum at one anchor is only a complete nested flag in a k_ℓ -dimensional coefficient space, then the flag-*

lattice formulation supplies at most

$$\sum_{j=0}^{k_\ell-2} (k_\ell - 1 - j) = \binom{k_\ell}{2} = B \quad (29)$$

independent linear incidence conditions at that anchor. Merely computing higher Hasse derivatives cannot replace B by $\binom{k_\ell}{3}$ once the complete flag has saturated.

Proof. At derivative order j , membership in the $(j+1)$ -dimensional flag subspace has codimension $k_\ell - 1 - j$. Summing these codimensions gives (29), equivalently the dimension of the complete flag variety. Changes of local parameter act lower-triangularly on the jet sequence and preserve only these nested spans. Additional conditions therefore require calibrated jets, transition data, or another invariant beyond the flag itself.

This rules out the tempting shortcut of assigning $\binom{k_\ell}{3}$ conditions to an “order-three relation” without identifying new public information. A genuine cubic improvement is not logically impossible, but it must break or quotient the lower-triangular reparametrization gauge; it cannot come from extending the complete nested flag recovered by Vedenev’s higher-order systems by one more derivative level. Compare the corresponding flag-dimension count in Vedenev’s Remark 6 [12].

There are three relevant solve models. Fresh dense elimination costs approximately U^3 field operations. A depth-first prefix tree shares the first $c-1$ intersections; if every prefix has the generic dimension, its most expensive intersection costs approximately

$$C_{\text{nest}} = B(U - (c-1)B)^2 \quad (30)$$

field operations. Finally, a suitable approximant-basis formulation would have leading term

$$\tilde{O}(k_\ell^{\omega-1}\sigma), \quad \sigma = cB, \quad (31)$$

but no reduction from the flag-lattice intersection to an approximant-basis instance with this cost has been derived. Giorgi and Lebreton use order bases inside block Wiedemann [7]; that result does not establish the flag-lattice reduction. Every order-basis row below is therefore an illustrative leading-monomial sensitivity, not a supported attack algorithm.

Proposition 3 (Bundle criterion). *View (27) as a rank- k_ℓ bundle on $\mathbb{P}^1$ with splitting $\mathcal{L} \simeq \bigoplus_j \mathcal{O}(-e_j)$. Then*

$$\sum_j e_j = cB, \quad \dim \mathcal{L}_{\leq D_\ell} = \sum_j \max(D_\ell - e_j + 1, 0). \quad (32)$$

In particular, if one $e_j = D_\ell$ and all remaining $e_j \geq D_\ell + 1$, then the correct solution space is one-dimensional.

Proof. Each anchor contributes determinant degree B , giving the first identity. The second is the dimension of the degree-bounded sections of the split line bundles on $\mathbb{P}^1$. The final assertion follows term by term.

Assumption 3 (Balanced planted splitting) *For flags arising from binary Goppa data, the lattice (27) has one $e_j = D_\ell$, every other $e_j \geq D_\ell + 1$, and prefix intersections of the generic dimensions used in (30).*

This single hypothesis supplies cross-anchor independence, final uniqueness, prefix-tree dimensions, and the output-degree profile needed by an approximant-basis implementation. For the $c = 8$, $k_\ell = 215$ cell the required profile has one $e_j = 854$ and the other 214 entries average 856.0093; for $c = 7$, $k_\ell = 258$, it has one $e_j = 897$ and the other 257 average 899.5097. These are required profiles, not measurements: no Popov basis has been computed for either shape.

Vedenev’s enumeration fixes two anchors by affine symmetry and guesses $c-2$ support coordinates and $c-1$ relative Frobenius phases [12]. With $q = 2^m$, its guess count is

$$G_{\text{aff}} = q^{c-2} m^{c-1}, \quad \log_2 G_{\text{aff}} = m(c-2) + (c-1) \log_2 m. \quad (33)$$

GRS covariance under $\sigma(z) = (\lambda z + \mu)/(\gamma z + \delta)$ permits three projective anchors to be normalized to $0, 1, \infty$, reducing the support factor to q^{c-3} . This is the action of $\text{PGL}_2(\mathbb{F}_q)$, abbreviated PGL below. Indeed, homogenizing a degree- $< r$ polynomial gives

$$\text{GRS}_r(\alpha, \beta) = \text{GRS}_r(\sigma(\alpha), \beta(\gamma\alpha + \delta)^{r-1}). \quad (34)$$

This proves the ambient code’s covariance. Carrying an infinite support point and the Frobenius branches through Vedenev’s Step 4 is not yet verified, so the extra factor- q saving is conditional.

Algorithm 5 makes the whole route explicit. Sidelnikov–Shestakov reconstruction and Hemmert’s Goppa-polynomial completion supply the final key representation [8,11]. Only full public-code equality authorizes an output.

If G denotes (33), the dominant downstream tally attached to Algorithm 5 is exactly Gm^2C_{solve} in the chosen model. A candidate that reaches the last line is first checked by multiplying the reconstructed $mt \times n$ parity check by the transpose of the public $k \times n$ generator and requiring the zero product; its rank is then checked separately. The cost is

$$C_{\text{keycheck}} \leq 2mtnk + O((mt)^2n) \quad (35)$$

binary gates. For `mceliece348864`, the leading product is $2^{33.8}$ gates, below every displayed per-guess solve; moreover, it is paid only for surviving candidates.

5.1 Optional Frobenius-Phase Synchronization

The phase factor m^{c-1} is a large but potentially removable part of (33). At anchor i , put $V_i = T_{0,i}/\langle y_i \rangle$. Under the rank condition checked in Algorithm 3, Vedenev’s Proposition 6 identifies the common nonzero zeros of the public quadratic

Algorithm 5: RecoverGoppaKey() : conditional end-to-end method.

- Input:** A public McEliece generator $\mathbf{G}'$, parameters (n, m, t) , a selected cell, and declared optimization flags.
- Output:** An equivalent Goppa key, or $\perp$.
- 1 Put the selected shortening in systematic form and choose c public hold-out positions;
 - 2 For each one-position hold-out, call **SupplyRelations** and **CertifyFirstOrder**; for every returned plane call **RecoverHigherFlag**; conditionally on Assumption 1, use an admissible multi-position hold-out cover instead;
 - 3 Normalize two anchors affinely; if the projective option is enabled, work homogeneously and normalize three to 0, 1, ∞ ;
 - 4 If phase synchronization is enabled, call **SynchronizePhases**; otherwise enumerate all relative Frobenius phases;
 - 5 Traverse the remaining support guesses depth first, sharing every prefix lattice intersection;
 - 6 At each leaf, solve (27); continue unless its degree- D_ℓ solution space is one-dimensional;
 - 7 Apply Sidelnikov–Shestakov to the exposed GRS code and Hemmert’s completion to obtain a candidate Goppa polynomial and support;
 - 8 Reconstruct a parity-check matrix H_{rec} ; verify $H_{\text{rec}}(\mathbf{G}')^\top = 0$ and separately verify $\text{rank } H_{\text{rec}} = mt$;
 - 9 **return** the first candidate passing exact equality, or $\perp$;
-

image in $\text{Sym}^2(V_i)^*$ with the m pure Frobenius lines [12]. Thus, conditional on the binary-Goppa hypothesis in Vedenov’s Conjecture 4, the unordered lines and their binary field algebra $\mathcal{E}_i \simeq \mathbb{F}_{2^m}$ are public; their missing cyclic label is exactly the relative Frobenius phase.

Fix a public irreducible polynomial of degree m and choose in each $\mathcal{E}_i$ one root A_i ; the m possible choices differ by Frobenius. Write $A_j^{[r]}$ for the endomorphism corresponding to the 2^r -th field power of that root, and define

$$\text{Sol}_{ij}(r) = \{C \in V_i^* \otimes V_j^* : A_i^\top C = C A_j^{[r]}\}, \quad r \in \mathbb{Z}/m\mathbb{Z}. \quad (36)$$

The following is the classical twist-space decomposition of $\text{Hom}_{\mathbb{F}_2}(E, E)$ by linearized polynomials. We include its short proof to fix the conventions used by the balance test, not as a novelty claim.

Lemma 3 (Frobenius-type decomposition). *If $\mathcal{E}_i$ and $\mathcal{E}_j$ are two binary regular representations of $\mathbb{F}_{2^m}$, then every $\text{Sol}_{ij}(r)$ has binary dimension m and*

$$V_i^* \otimes V_j^* = \bigoplus_{r \in \mathbb{Z}/m\mathbb{Z}} \text{Sol}_{ij}(r). \quad (37)$$

Proof. Identify both modules with $E = \mathbb{F}_{2^m}$. The trace pairing identifies a bilinear form with a unique binary-linear map $L : E \rightarrow E$ by

$$C_L(x, y) = \text{Tr}_{E/\mathbb{F}_2}(xL(y)), \quad L(y) = \sum_{s=0}^{m-1} a_s y^{2^s}.$$

If the common generator is α , the balance relation is $L(\alpha^{2^r}y) = \alpha L(y)$; it selects $s \equiv -r \pmod{m}$, while $a_s \in E$ is free. The m selected coefficient spaces therefore have dimension m , are independent, and exhaust all m^2 binary-linear maps.

A synthetic experiment tested (36) after independent random binary basis changes at eight local field algebras. Across 24 trials at every $m \in \{4, 6, 8, 12\}$, the true shift was unique, the adjacent wrong shift was rejected, and (37) held. A random pairing was pure, meaning it was contained in one summand, in zero of 192 draws at each degree. At $m = 12$, at most six sampled pairs (x, y) determined the shift, with mean 4.25. The solution-space dimension was m for every r , so dimension does not discriminate; membership of the specific pairing does. This is heuristic evidence from synthetic field algebras, not evidence that the required pairing can be constructed from public hold-out relations, and the public artifact does not reproduce the experiment.

The remaining type obstruction is decisive. For the two-position relation space K_{ij} , the canonical order-two jet takes values in

$$K_{ij} \longrightarrow \mathrm{Sym}^2(V_i)^* \oplus \mathrm{Sym}^2(V_j)^*, \quad (38)$$

not in the mixed space $V_i^* \otimes V_j^*$. Separate jets at two base points provide no canonical cross block. Frobenius equivariance alone would not suffice in any event: by Lemma 3, a generic equivariant-looking pairing has components in several summands.

Assumption 4 (Public pure cross-pairing) *For every edge (i, j) of a spanning tree, a public fail-closed algorithm `CrossPair` maps exactly r_{pair} certified K_{ij} relations and the two certified local spaces to a nonzero $C_{ij} \in V_i^* \otimes V_j^*$ lying in exactly one $\mathrm{Sol}_{ij}(r)$. The displayed rows assume $r_{\mathrm{pair}} = r_{\min}$ and that its arithmetic after relation supply is lower-order than the corresponding two-position relation-generation computation; any larger required block changes the batch count explicitly.*

Ambient second polars based at y_i , y_j , or $y_i + y_j$ are natural candidates for `CrossPair`, but none is proved to satisfy Assumption 4. The $h = 2$ relation supply itself is dimension-admitted at the two priced cells: its kernel floors are $5.54 \cdot 10^{12}$ for $(c, k_\ell, d) = (8, 215, 8)$ and $3.06 \cdot 10^{15}$ for $(7, 258, 9)$. Thus, the missing object is the public map, not the existence of a target-scale relation.

Let $\bar{\rho}_2$ and ν_2 denote the row-rank bound and nonzero count for the $h = 2$ cell, and put $a_{\mathrm{pair}, b} = \lceil r_{\mathrm{pair}}/b \rceil$. Under the conservative width- b charging with $r_{\mathrm{BW}} = 1$, the dominant additional work in Algorithm 6 is

$$C_{\mathrm{pair-sup}} = (c - 1) 6 \cdot 64 \left\lceil \frac{2\bar{\rho}_2}{b} \right\rceil \nu_2 a_{\mathrm{pair}, b}. \quad (39)$$

The displayed cost takes E to be a spanning tree; optional non-tree cycle checks add one two-position relation-generation computation per checked edge. The small-field decomposition and consistency arithmetic are lower-order at $m \leq 13$. Equation (39) prices the relation supply, not a proof of Assumption 4; every phase-aligned number therefore remains conditional even when this cost is included.

Algorithm 6: SynchronizePhases() : conditional singleton-space synchronization.

- Input:** Certified singleton relation spaces and local flags at anchors $1, \dots, c$; admissible $h = 2$ cells; declared r_{pair} .
- Output:** Relative phases $e_i \in \mathbb{Z}/m\mathbb{Z}$, or $\perp$.
- 1 For each anchor i , recover the unordered pure lines from its certified quadratic image; construct $\mathcal{E}_i$ and refuse unless it is a degree- m field with generator A_i ;
 - 2 Choose a connected edge set E on the anchors and, for each $(i, j) \in E$, call **SupplyRelations** with held set $\{i, j\}$ to obtain r_{pair} certified relations $K'_{ij} \subseteq K_{ij}$;
 - 3 For each $(i, j) \in E$, set $C_{ij} \leftarrow \text{CrossPair}(K'_{ij}, V_i, V_j)$ as required by Assumption 4; refuse if the public checks fail;
 - 4 Decompose every C_{ij} using (37); refuse unless exactly one component is nonzero, and record its index r_{ij} ;
 - 5 Assign phases along a spanning tree and verify $r_{ij} = e_j - e_i$ on every remaining edge;
 - 6 **return** $(e_1, \dots, e_c)$ modulo one global shift;
-

6 Cost Model

We express all work as classical Boolean gates. Following NIST’s PQC evaluation criteria, Categories 1, 3, and 5 are compared with 2^{143} , 2^{207} , and 2^{272} gates, respectively; 2^{128} is a number of AES-128 key checks, not the corresponding gate count.* One width-64 XOR word operation is charged as 64 gates. One $\mathbb{F}_{2^m}$ multiplication is represented by the optimistic m^2 AND-gate proxy; multiplier XORs are omitted. A 250–300 gate sensitivity at $m = 12$ would add 0.80–1.06 bits to downstream terms, but not to the binary supplier. These are comparison units, not hardware measurements; in particular, random memory access, address generation, communication, and storage are uncharged.

For a declared solve model, let C_{solve} be one of U^3 , $B(U - (c - 1)B)^2$, or the leading monomial $k_\ell^{\omega-1}cB$, measured in $\mathbb{F}_q$ operations. Let $f = 2$ for affine normalization and $f = 3$ for the conditional projective normalization. Without phase synchronization, the downstream work is modeled as

$$C_{\text{down}} = 2^{m(c-f)} m^{c-1} m^2 C_{\text{solve}}. \quad (40)$$

Under Assumption 4, Algorithm 6 deletes the factor m^{c-1} . Post-solve matching, GRS reconstruction, Goppa completion, and full verification are lower order in the displayed shapes. The one-time higher-flag arithmetic is charged by (26). The total is the sum

$$C_{\text{total}} = C_{\text{sup}} + C_{\text{flag}} + C_{\text{down}} + C_{\text{align}}. \quad (41)$$

* [https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/evaluation-criteria/security-\(evaluation-criteria\)](https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/evaluation-criteria/security-(evaluation-criteria))

For an uncharged phase-aligned subtotal, C_{align} is absent rather than zero. For a subtotal that charges pairwise relation generation, the model substitutes (39); this accounts for the proposed $h = 2$ relation-supply computations but remains conditional on Assumption 4 and its lower-order post-supply clause. Every row additionally uses one unconditioned binary Coppersmith attempt and Assumption 2; no displayed exponent is an unconditional expected running time.

The complete optimization variables are

$$(d, k_\ell, h, \{s_p\}, \{J_p\}, c, b, \text{held-set use, normalization, solve model}). \quad (42)$$

Optimizing only the sparse supplier, holding c fixed, or pinning $b = 64$ gives different and generally inferior points.

7 Concrete Tally

7.1 Selected Supplier Cells

Table 2 gives the heterogeneous cells that minimize modeled relation-generation cost in an exhaustive $h = 1$ search over $3 \leq d \leq 24$ and every admissible k_ℓ , at width 64 and projection weight $\lfloor k_\ell/2 \rfloor$. The displayed supplier cost includes one nominal Coppersmith attempt for each required width-64 relation batch on the exact augmented operator, all one-position hold-outs, and conversion to gates. It does not include downstream work or an expected retry multiplier. All five rows have $d = 8$, high/low Lucas sets $\{0, 4, 6\}/\{0, 4, 5\}$, and Hermite slack one.

Table 2. Selected relation-generation cells for the five Classic McEliece parameter sets. Work columns are base-two logarithms of gates with $r_{\text{BW}} = 1$; state is the width-64 vector panel plus safe-rank-bound sequence from (22).

Parameter Set	k_ℓ	Schedule	c	$\log_2 N$	$\log_2 C_{\text{sup}}$	$\log_2$ state bits
mceliece348864	208	719@7 + 48@6	9	46.11	114.35	54.69
mceliece460896	269	1228@7 + 19@6	10	49.12	121.49	57.70
mceliece6688128	305	1584@7 + 79@6	12	50.59	125.11	59.17
mceliece6960119	295	1484@7 + 62@6	11	50.20	124.10	58.78
mceliece8192128	305	1584@7 + 79@6	12	50.59	125.11	59.17

Published hold-out baseline. In the parameter-set order of Table 2, Ghoshal et al. use uniform order $s = 7$ and report [6]

$$\begin{aligned} k_\ell &= (217, 274, 315, 304, 315), \\ \log_2 T_{\text{atk}} &= (114.16, 120.29, 123.95, 123.02, 123.95), \\ \log_2 S_{\text{atk}} &= (66.45, 69.54, 71.37, 70.91, 71.37). \end{aligned}$$

These numbers price a different output: T_{atk} is one reliable small-field block-Lanczos consistency test $M_\tau x = b_\tau$ at one held column, whereas C_{sup} asks the augmented solver for r_{min} verified right-kernel relations at each of c anchors. For `mceliece348864`, our five width-64 batches at nine anchors multiply the one-anchor, one-batch charge by 45, and $114.3533 - \log_2 45 = 108.8615$. Thus the proximity of our 114.35 to their 114.16 conceals different output and solver models rather than establishing a slower or faster like-for-like supplier. The storage-exponent gap, $66.45 - 54.69 = 11.76$ bits, is likewise not a direct improvement: their S_{atk} includes the stored sparse system and the state of a conditioned reliable solver, whereas our S_b retains only the width-64 vector panel and sequence under matrix-free generation and excludes conditioning, I/O, and temporary solver memory. The comparison nevertheless isolates the intended memory simplification—matrix-free generation with a narrow retained block—but whether the 11.76-bit gap survives reliable conditioning and implementation is open.

The schedules are not cosmetic: every corresponding uniform schedule is refused. The lower adjacent uniform order misses Hermite, while the higher order has a row-rank bound exceeding N . The heterogeneous schedules occupy the intervening one-unit window. The artifact model reproduces the exact search and every row of Table 2.

Hermite slack one is an exact algebraic threshold, not a probabilistic success margin. It is nevertheless useful to ask whether the selected cell depends numerically on equality at one unit. As Table 3 shows, assigning order seven to every constrained column at the $c = 8$, $k_\ell = 215$, $d = 8$ cell buys the maximum admissible slack 42 for only 0.078 bits in the relation-generation exponent; slack 43 would require order eight and is refused by the $s_{\text{hi}} \leq d - 1$ gate. Thus the one-unit choice is immaterial to the cost estimate, although additional slack does not strengthen a correctly applied Lemma 1.

Table 3. Hermite-slack sensitivity at $(k_\ell, d, h, b) = (215, 8, 1, 64)$. Work is $\log_2$ supplier gates.

Slack	Schedule	Kernel floor	Supplier	Δ
1	726@7 + 41@6	6,279,785,122,167	114.86	+0.000
42	767@7 + 0@6	1,187,277,582,725	114.93	+0.078

7.2 All Classic McEliece Sets

Table 4 adds the downstream solve and the one-time bound (26) to the cells of Table 2. We write “affine+dense” for Vedenev’s two-anchor normalization with a fresh U^3 solve per guess, and “PGL+nested” for the unverified third normalized anchor with Assumption 3; the prefix names the normalization and the suffix the solve model. Every total still assumes Assumption 2, successful yield from one binary Coppersmith attempt per required relation batch, and correct global flag-lattice/GRS reconstruction for binary Goppa data.

Table 4. Modeled complete work at the selected supplier cells. Entries are $\log_2$ classical gates. Values below a comparator are not established attacks; the shared missing hypotheses are listed in Section A.

Parameter Set	PQ Cat / target	Supplier cost	Affine + dense	PGL + nested	State $\log_2$ bits
mceliece348864	1 / 143	114.35	172.13	146.29	54.69
mceliece460896	3 / 207	121.49	200.03	176.77	57.70
mceliece6688128	5 / 272	125.11	235.09	207.83	59.17
mceliece6960119	5 / 272	124.10	217.96	194.35	58.78
mceliece8192128	5 / 272	125.11	235.09	207.83	59.17

The Category 3 and 5 rows fall below their gate comparators already in the affine+dense model, but this does not establish breaks: those rows inherit the unproved binary reconstruction, cross-anchor rank, operator backend, uniform-weight nonzero model, and uncharged memory. Their cells were selected for supplier cost rather than end-to-end optimality, so the table is a reproducible tally rather than a claim of global minima.

7.3 The Category 1 Pareto Surface

For mceliece348864, changing the cell moves the anchor count and dominates the downstream. The relevant ladder is shown in Table 5; its width-64 state grows much faster than its arithmetic falls.

At $c = 7$, $U = 231,684$ and the affine guess count is $2^{81.51}$. A dense solve costs $2^{60.64}$ gates per guess, giving $2^{142.15}$ total, only 0.85 bits below the Category 1 comparator. This is the least structurally aggressive below-target model: it does not require PGL normalization, prefix nesting, an order-basis formulation, multi-position hold-out richness, or phase alignment. It still requires Assumption 2, successful binary Krylov yield under $r_{BW} = 1$, correct global binary reconstruction, independent anchor conditions with only a 0.167% surplus, and a complete relation-generation implementation. Reducing the block width to $b = 32, 16, 8$ gives respectively $2^{60.7}$, $2^{59.7}$, and $2^{58.7}$ bits of retained state; conservative supplier work rises by one, two, and three bits, but remains below the dense downstream, so the rounded total stays $2^{142.15}$.

Table 5. mceliece348864 anchor ladder at width 64 with $r_{\text{BW}} = 1$ and one-position relation generation. “Affine+dense” and “PGL+nested” are the two variants defined above.

Hold-outs c	Config (k_ℓ, d)	Supplier cost	Guesses $(\log_2)$	Affine + dense	PGL + nested	State $\log_2$ bits
9	(208, 8)	114.35	112.68	172.13	146.29	54.69
8	(215, 8)	114.86	97.09	156.73	135.71	55.00
7	(258, 9)	128.80	81.51	142.15	128.81	61.66
6	(322, 9)	134.15	65.92	134.16	134.15	64.18
5	(429, 10)	152.60	50.34	152.60	152.60	72.96

There is a useful downstream-independent limit. Exhaustively minimizing C_{sup} over all 19,338 dimension-admitted $h = 1$ cells with $3 \leq d \leq 24$, width 64, projection weight $\lfloor k_\ell/2 \rfloor$, and the direct Lucas traversal returns the original $(k_\ell, d, c) = (208, 8, 9)$ cell at $2^{114.3533}$ gates for one nominal attempt per required relation batch. Since (41) is a sum of nonnegative terms, no change confined to normalization, phase handling, Step 3, or final reconstruction can put this fixed one-position, $r_{\text{BW}} = 1$ direct-traversal model below that supplier floor. This is a lower bound on the modeled attack cost, not an upper complexity bound or a reliable expected-work bound; multi-position hold-out covers, a systematic basis with lower non-pivot weights, operator elimination, retries, or a different relation-supply representation change the model and can cross it.

Table 6. Exhaustive fixed-model supplier floor for mceliece348864. The tuple is (k_ℓ, d, c, b) .

Scope	Admitted cells	Minimizer	Supplier	$\log_2$ state bits
Singleton, $w = \lfloor k_\ell/2 \rfloor$	19,338	(208, 8, 9, 64)	114.35	54.69

With phase alignment disabled, an exhaustive search over all 19,338 admitted singleton cells with $3 \leq d \leq 24$, every supported width $b \in \{1, 2, 4, 8, 16, 32, 64\}$, and every admissible common hold-out variant gives Table 7.

Table 7. Exhaustive mceliece348864 optima for the PGL+nested variant under state-size caps, all supported widths, and $r_{\text{BW}} = 1$. Cells are shown as (c, k_ℓ, d) . Common-kernel relation generation, marked † , additionally assumes common-kernel flag richness.

$\log_2$ cap	Cell	Width	Hold-out use	Total	$\log_2$ state
53.00	(8, 256, 8)	4	singleton	127.70	52.70
56.32	(8, 256, 8)	32	singleton	126.89	55.70
59.64	(8, 256, 8)	64	singleton	126.88	56.70
63.00	(7, 258, 9)	64	common †	126.13	61.73

The $k_\ell = 256$, $d = 8$ cell is the last admitted cell at that degree. Its surviving dimension at the costliest nested-intersection step is only 896 of $U = 229,376$, or 0.39%, so all capped rows place Assumption 3’s generic-prefix hypothesis exactly where it does the most work. The caps apply to S_b only: streamed relation blocks, temporary memory for generator-basis computation, and access costs are outside the model. These are exhaustive outputs of the declared model, not recommended configurations or results from executed attacks.

First, Table 8 isolates the computational benefit of phase alignment by omitting C_{align} in (41). These are not totals. At width 64, the best dense and nested models use $c = 8$, while the illustrative order-basis leading monomial favors $c = 9$. Under a 2^{53} -bit retained-state cap, $c = 8$ with nesting requires width 8 and costs $2^{120.62}$, whereas the $c = 9$ illustrative order-basis row fits at width 16 and costs $2^{118.14}$. Their 2.48-bit separation is still smaller than the uncertainty introduced by the missing flag-lattice reduction and solver constants; the table compares sensitivities, not order-basis claims.

Table 8. Conditional phase-aligned subtotals for `mceliece348864` with $r_{\text{BW}} = 1$. Public construction and execution of synchronization are omitted; *order-basis rows are illustrative.

Cell and solve	Guess count	Downstream	Supplier	Subtotal	$\log_2$ state
$c = 8$, $b = 8$, PGL+nested	60.00	110.61	120.62	120.62	52.00
$c = 9$, $b = 16$, PGL+order basis*	72.00	112.13	118.12	118.14	52.69
$c = 8$, $b = 32$, PGL+nested	60.00	110.61	116.71	116.73	54.00
$c = 9$, $b = 32$, PGL+order basis*	72.00	112.13	116.20	116.28	53.69
$c = 8$, $b = 64$, PGL+dense	60.00	119.63	114.86	119.69	55.00
$c = 8$, $b = 64$, PGL+nested	60.00	110.61	114.86	114.93	55.00
$c = 9$, $b = 64$, PGL+order basis*	72.00	112.13	114.35	114.63	54.69

Table 9 then adds exactly $c-1$ two-position relation-generation computations using (39) and $r_{\text{pair}} = r_{\text{min}}$. It assumes that `CrossPair` exists and that its post-supply work is lower-order, so “priced subtotal” means cost-complete only under Assumption 4, not an established attack. At $c = 8$, width 32, the extra supply raises $2^{116.73}$ to $2^{117.63}$, an increment of only 0.90 bits, while eliminating 25.09 bits of phase enumeration relative to the no-alignment row. At $c = 7$ relation generation already dominates, so buying synchronization does not improve the total.

Table 9. Conditional cost of phase alignment for `mceliece348864`. “Omitted” deletes phase guesses without charging alignment. “Pair supply” charges $c-1$ admissible $h = 2$ relation-supply computations. Every row using alignment remains conditional on Assumption 4.

Cell	b	No sync.	Omitted	Pair supply	Subtotal	Added bits	$\log_2$ state
$c = 8$, $k_\ell = 215$	64	135.71	114.93	114.67	115.81	0.88	55.01
$c = 8$, $k_\ell = 215$	32	135.71	116.73	116.52	117.63	0.90	54.01
$c = 8$, $k_\ell = 215$	16	135.71	118.63	118.44	119.54	0.91	53.01
$c = 7$, $k_\ell = 258$	32	130.80	130.80	130.59	131.70	0.90	60.67

8 Potential Reductions in Relation-Generation Cost

8.1 Systematic-Basis Weight

The supplier floor in Table 6 uses weight $\lfloor k_\ell/2 \rfloor$ for every non-pivot column. The exact nonzero formula (14) is strongly weight-dependent even though the row-rank charge is stable above weight $d + s - 1$. Holding the $c = 8$, $k_\ell = 215$, $d = 8$ cell and schedule fixed gives Table 10.

Table 10. Direct-traversal nonzero sensitivity to a uniform non-pivot weight at the $c = 8$ cell. Δ is the supplier-exponent change from weight 107.

Weight	Nonzeros	Δ
60	190,035,290,622,323,575	-2.15
75	327,641,960,639,229,275	-1.36
90	525,011,004,642,895,485	-0.68
107	843,010,975,919,840,351	+0.00
120	1,171,737,009,839,400,485	+0.48
140	1,861,805,299,521,543,285	+1.14

The attacker may choose a shortening and a systematic basis, but cannot prescribe the resulting fundamental-circuit weights independently. Finding a systematic basis with lower non-pivot weights is a public matroid-basis optimization problem, plausibly attacked by ISD-style basis sampling or local basis exchanges. Before adding any savings in Table 10 to a total, we must measure its preprocessing cost and attainable weight distribution.

Two generic shortcuts do not yet earn a modeled factor. First, a direct restricted subset-zeta evaluation costs about wN operations per constrained point. At $w = 107$ this is 9.66 times the average Lucas-compressed sparse traversal $\nu/(mt - h)$, because the transform computes derivative levels that Lucas compression has already removed. Second, sparse structured elimination may help, as in integer-factorization matrices, but the present operator has average column degree 8,493 under the $c = 8$ uniform-weight convention. Without degree histograms and fill measurements at the target cell, importing a generic 5–20-fold dimension reduction would be unsupported.

8.2 Rank-Only Queries

Some local gates ask only for image dimensions. For any public jet block J ,

$$\text{rank} \begin{pmatrix} M \\ J \end{pmatrix} - \text{rank } M = \text{rank}(J|_{\ker M}), \quad (43)$$

and successive differences similarly give the rank of a quadratic jet restricted to the kernel of the linear jet. Black-box rank estimation can therefore replace explicit relation vectors for scalar corank tests. It does not by itself construct

the image bases, zero-scheme certificate, local flags, or relations required by the global reconstruction. The frequently suggested savings of all five relation batches plus the augmented factor two is consequently unpriced: one must first specify the designated rank or membership queries, their count, and their reliability over $\mathbb{F}_2$.

9 Asymptotic Reading

The supplier feasibility window compares the dimension requirement $k_\ell \gtrsim d\sqrt{mt}$ with the Hermite ceiling $k_\ell \lesssim d(2t+1) - mt$. They become compatible around $d > m/2$. At the minimal supplier scale,

$$\log_2 N \sim \frac{m}{2} \left(\log_2 e + \frac{1}{2} \log_2(mt) \right), \quad \log_2 C_{\text{sup}} \sim m \left(\log_2 e + \frac{1}{2} \log_2(mt) \right). \quad (44)$$

When $m \asymp \log_2 n$, this is quasi-polynomial, approximately $2^{\frac{1}{2}(\log_2 n)^2}$ up to constants. It is only the supplier complexity. At the same scale $c \asymp 2\sqrt{mt}/d$; if, in addition, $mt = \Theta(n)$, then the support and phase enumeration is $2^{\Theta(\sqrt{n})}$. Without that fixed-codimension-fraction regime, the exponent should be left as a function of mt . End-to-end quasi-polynomial key recovery therefore requires an additional downstream shortcut, a different shortening scale, or both. Conditional on Assumption 4, the phase synchronizer of Algorithm 6 removes only the m^{c-1} factor, not the support factor q^{c-f} .

10 Conclusion

The hold-out key-recovery route can be described without hiding any large step behind a single “polynomial” label. Mixed multiplicity creates admissible binary kernels that uniform multiplicity misses; Lucas compression preserves their row spaces; the augmented black-box identity isolates the correct right kernel; sequential higher-flag recovery is exposed as a separate conjectural interface; and every downstream guess solves an explicit U -unknown flag-lattice problem before GRS and Goppa reconstruction. These pieces yield a reproducible conditional arithmetic tally for all Classic McEliece parameter sets and a compute-memory surface for `mceliece348864`.

The accounting identifies the research priorities. The immediate relation-generation problem is a reliable matrix-free binary kernel algorithm with explicit conditioning, success probability, retry policy, and memory-access cost. Separately, a generated binary-Goppa experiment must distinguish first-order success from complete higher-flag recovery using the same truncated block priced here. A public basis search can test whether systematic bases with lower non-pivot weights reduce relation-generation cost, and multi-position hold-out flag tests can decide whether the exact $h = 4$ cover saving is real. A theorem constructing a nonzero pure cross-pairing from public two-position relations would turn the synthetic phase test into an attack component. Until these links close, the results are a costed conditional attack program rather than a break of Classic McEliece.

On the Use of AI in Cryptanalysis

These results originate from a systematic effort to implement and improve cryptanalytic attacks on Post-Quantum Cryptography using AI. We used OpenAI GPT-5.6-Sol and Claude Opus 5 for theory formulation, proofs, analysis, and preparing and executing experiments. Furthermore, we used the LLMs and Grammarly to prepare and improve the presentation. The process was closely human-supervised, however, requiring hundreds of prompts and manual edits.

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A Cost-Estimate Component Status

In the following, we collect the proof, modeling, and conjectural status of every ingredient used in our cost estimation.

Used in	Ingredient and status
§3.1; Alg. 1	Hermite soundness and heterogeneous schedule. Proved in Lemma 1 and Proposition 1; cell arithmetic exact.
§3.2; Alg. 1	Lucas-minimal levels. Proved in Theorem 1 from the binary inclusion-matrix decomposition and reproduced arithmetically by the artifact model.
§3.3; Alg. 1	Row-rank bound and kernel floor. Exact point-block upper bound and rigorous kernel lower bound; no cross-point rank equality claimed.
§3.3; Algs. 1–2	Nonzero count. Exact for the chosen public-column weights; target model uses the uniform-weight convention rather than an opened key.
§4; Alg. 2	Augmented kernel identity. Exact by Lemma 2. The tables price one nominal equal-block Coppersmith attempt per relation batch on the rank- $2\bar{\rho}$ augmented matrix; exceptional projections, a binary preconditioner, failure probability, expected retries, and a target-scale backend are open.
§4; Alg. 5	Multi-position hold-out cover. Containment, Hermite re-scheduling, and direct costs exact; recovery of every singleton flag from enlarged kernels is open Assumption 1.
§5; Alg. 5	Anchor count and final uniqueness. Depend on Assumption 3; no binary target-scale Popov profile measured.
§5; Alg. 5	Per-anchor condition count. $\binom{k_\ell}{2}$ is exact for complete flag data by Proposition 2; more conditions require a new invariant, not merely a higher derivative level.
§5; Alg. 5	Prefix-tree nesting. Algorithmically explicit; savings depend on the prefix dimensions in Assumption 3.
§5; Alg. 5	PGL₂ normalization. Ambient GRS covariance exact; integrating $0, 1, \infty$ into Step 4 branch bookkeeping remains open.
§5; §6	Order-basis cost. Leading soft- O model on an unworked target-scale formulation.

Continued on next page

Cost-Estimate Component Status (continued)

Used in	Ingredient and status
§5; Algs. 3, 5	First-order branch planes. Public quadratic forms and rank tests are explicit; their identification with the true binary-Goppa Frobenius lines depends on Vedenev’s Conjecture 4.
§5; Algs. 4–5	Sequential higher flags. Algorithm 4 exposes Vedenev’s systems (20)–(21). Correctness depends separately on Conjecture 3 and on Assumption 2’s truncated-block richness; dense arithmetic and a declared branch multiplier are priced.
§5.1; Alg. 6	Frobenius-type decomposition. Classical linearized-polynomial decomposition, reproved in Lemma 3 to fix conventions; a synthetic experiment gives heuristic support to the balance test and non-genericity of purity but is not reproduced by the artifact.
§5.1; Alg. 6	Public pure cross-pairing. Open Assumption 4. The canonical $h = 2$ jet has no mixed block. The $c - 1$ two-position relation-generation computations are priced in Table 9, but the map they should feed is not constructed.
§5; Alg. 5	Global binary reconstruction. Conjectural after local flags; Vedenev reports end-to-end experiments only over $\mathbb{F}_4$ [12].
§4; §6; Alg. 2	Memory. Vector panel and safe-rank-bound Coppersmith sequence counted by (22); relation streaming, temporary memory for generator-basis computation, access, communication, replication, and NIST memory penalties remain unpriced.
§5; Alg. 5	Final key verification. Unconditional once a candidate is produced: reconstruct the complete Goppa parity-check row space and compare it with the public dual.

Accordingly, none of Tables 4, 5 and 7 to 9 establishes a key recovery at its displayed exponent. The strongest narrow statement is that, if one nominal binary Coppersmith attempt per required relation batch yields the required relations, Assumption 2 holds for the truncated blocks, global binary reconstruction is correct, and the seven-anchor lattice has the required rank, then the explicitly described $c = 7$ route has a $2^{142.15}$ modeled gate count before memory penalties and $2^{61.7}$ bits of retained width-64 solver state, excluding temporary solver memory. In the fixed one-position, $r_{\text{BW}} = 1$ direct-traversal model, $2^{114.35}$ is the exhaustive supplier floor, so downstream work can approach it from above but cannot go below it. The phase-aligned rows omit or conditionally price synchronization; even the pair-priced $c = 8$ rows from $2^{115.81}$ to $2^{119.54}$ remain conditional because CrossPair is absent. Charging an unconstructed method is not constructing it.